THEALON AI ENGINEERING CHALLENGE

Podcast Companion: Research & Engineering Findings

1. Why We Built This (User Value)

Long podcasts have lots of useful information, but remembering or looking for particular information in them is difficult. Ordinary AI chatbots provide generic information that you cannot confirm.

Who benefits most:

- **Students & Learners:** Need to easily review fundamental concepts and assess understanding before exam.
- **Busy Listeners:** Wants to know everything in 2 minutes rather than listening or watching an 80 minute episode.
- **Readers & Accessibility:** People who can understand and remember information better through written words than voice.
- **Podcast Creators:** Need chapter markers and quiz questions to engage their listeners.

The Core Problem Solved:

The Core Problem Solved: Instead of guessing, we answer: "Where did the speaker actually say that, and can I trust it?" Every takeaway, chapter, and quiz answer links directly to the exact words in the podcast transcript.

2. Technical Tools Compared

We tested multiple free tools to find the best combination for speed, reliability, and cost.

A. AI Text Models (Summaries, Chapters, Quiz)

Model	Key Strengths	Limitations	Our Verdict
Google Gemini Flash	Huge context window: Reads 10+ hours in one call. Native JSON support. Generous free tier.	Free tier can experience queue delays during peak traffic hours.	Primary Choice (handles long episodes).
Groq (GPT-OSS-120B)	Super fast: Answers in 1 to 2 seconds. Reliable JSON mode.	Free tier has an 8,000 token per minute ceiling; cannot fit huge files.	Fast Backup & comparison model.
Hugging Face Router	Provides access to diverse open-source models with one API	Free tier gives small monthly credits that can run out quickly.	Secondary Fallback.

	key.		
Local Models (Ollama)	100% private with zero API rate limits.	Needs a powerful computer/GPU. Cannot run on free cloud servers (512MB RAM).	Rejected for free cloud hosting.

Real-World Finding: AI models retire quickly! While developing, Groq retired llama-3.3-70b-versatile. Because our system makes real API calls, we caught this immediately and upgraded to openai/gpt-oss-120b.

B. Speech-to-Text (Audio Transcribers)

Transcriber	**Key Strengths**	**Limitations**	**Our Verdict**
Groq Whisper	Real timestamps: Second-by-second accuracy via verbose_json. Over 200x faster than real-time.	25 MB file size limit on free tier.	Primary Choice (we compress and split audio into chunks).
Gemini Audio	Handles many spoken languages well natively.	Timestamps are not dependable; inline audio capped at 20 MB.	Backup Choice (drops timestamps honestly).
OpenAI Whisper API	High quality reference implementation.	Requires a paid credit card account.	Rejected (breaks free-tier rule).

C. Cloud Hosting Platforms

Platform	**Findings & Evaluation**	**Our Verdict**
Render (Free Web Service)	Runs Python and Docker smoothly without credit cards. Allows audio uploads without small file ceilings. Sleeps after 15 min idle.	Chosen Host (Live on Render).
Vercel	Has a strict 4.5 MB request	Rejected (cannot upload

	body limit. Real podcast audio files are 20–60 MB, so Vercel blocks them.	audio).
Hugging Face Spaces	Free Docker compute spaces now require paid accounts for new users.	Rejected (requires paid plan).

3. How We Keep Answers Trustworthy (Reliability)

Instead of hoping the AI doesn't hallucinate, our Python backend strictly checks everything:

- **Numbered Passages:** The transcript is divided into numbered passages (Passage 1, Passage 2, etc.). The AI has to include the passage number in its response.
- **Verbatim Quote Checks:** When generating quizzes and answering questions, the server does a search in the transcript to verify the quote exists verbatim. Ungrounded questions are ignored.
- **Honest Timestamps:** Timestamps should come from either Whisper output or subtitle file (.srt/.vtt). In case of a text file input, the application honestly outputs "No timestamps available".
- **Server-Side Option Shuffling:** Often AI model puts the right answer as option B. This is avoided through shuffling answers on the server side.
- **Abstaining on Out-of-Domain Questions:** In case the user asks an out of domain question ("Who won the World Cup?" in relation to the podcast), the system politely declines.

4. Safety & Security (OWASP Top 10 API Security)

- **File Header Checks (Magic Bytes):** We analyze the binary header of any file that gets uploaded. A text or harmful file renamed to .mp3 gets detected safely and rejected before being processed by ffmpeg.
- **Input Guardrails:** Transcripts shorter than 200 characters or longer than 150,000 characters get safely rejected with proper HTTP error codes.
- **Privacy Notice:** Users are warned beforehand that free tiers of AI use user inputs, and that users should not send passwords or confidential documents.
- **Secret Hygiene:** No passwords or API keys are hard coded anywhere in the code base or even on GitHub.

5. Measured Benchmark: Gemini vs. Groq Inside Our App

We ran both models head-to-head on the exact same 11-minute podcast transcript (about 10,300 characters). Both models were pinned with no hidden fallback:

Benchmark Metric	**Google Gemini Flash-Lite**	**Groq (GPT-OSS-120B)**
Overview Generation Time	5.2 seconds	1.5 seconds (over 3x faster)

Chapters Generation Time	1.5 seconds	1.8 seconds
Quiz Generation Time	5.8 seconds	3.5 seconds
Verbatim Quote Check	100% Verified (3 of 3 questions)	100% Verified (3 of 3 questions)
Off-Topic Question Refusal	Refused correctly (no guess)	Refused correctly (no guess)
Maximum Episode Length	Huge (handles 10+ hour audio)	Medium (limited by 8K tokens/min)

Takeaway: Groq is unmatched for speed (1.5 seconds), while Gemini is unmatched for episode length. Our Auto mode uses Groq for speed when possible, and smoothly relies on Gemini for long marathons.

6. Three Future Improvements (Next Steps)

- **1. Automated Truthfulness Testing (Highest Priority):** An automated benchmark harness that checks if every summary claim strictly matches the podcast audio without any subtle drift.
- **2. Durable Database & Shareable Links:** Storing episode companions in a lightweight database so users can share permanent links with friends.
- **3. Word-by-Word Streaming:** Streaming tokens over Server Sent Events (SSE) so users see summary sentences appear instantly as they are generated.

7. References & Research Sources

- **Edison Research — The UK Podcast Consumer 2025:** Official media study documenting podcast consumption behaviors (51% monthly listeners): https://www.edisonresearch.com/the-uk-podcast-consumer-2025/
- **OpenAI Speech-to-Text Documentation:** Audio transcription guide and best practices for Whisper implementations: https://platform.openai.com/docs/guides/speech-to-text
- **OWASP API Security Top 10:** Industry-standard API security recommendations for file uploads, auth, and input sanitization: https://owasp.org/www-project-api-security/
- **Groq Cloud Documentation:** Ultra-fast inference engine API references and rate limit specifications: https://console.groq.com/docs
- **Google AI Gemini API Documentation:** Context window architectures and multimodal API references: https://ai.google.dev/docs